

EDUCATION

McGill University & Mila - Quebec AI Institute <i>Master's of Science in Computer Science</i>	Montréal, QC <i>Aug. 2025 – Present</i>
Duke University / Duke Kunshan University <i>Bachelor's of Science in Data Science</i>	Durham, NC / Suzhou, China <i>Aug. 2021 – May 2025</i>

GRANTS & AWARDS

McGill CS Graduate Award — \$30,000/year	2025
McGill CS Graduate Registration Bonus — \$5,000	2025
DKU Student Experiential Learning Fellowship — \$1,000	2024
DKU Signature Work Research Grant — \$1,500	2023
DKU Innovation Incubator — \$3,000	2023

RESEARCH EXPERIENCE

Machine Learning Research Intern <i>University Mohammed VI Polytechnic</i> - Unified a 10-yr global crop-yield to a fertilizer dataset (N/P/K) across 150+ countries. - Shipped transformer field-recs and an agentic LLM + retrieval copilot (RAG) with 2-second lookup latency.	Jul. 2025 – Present <i>Benquerir, Morocco</i>
Undergraduate Research Assistant <i>Duke Kunshan University</i> - Built a 300k-image, ground-truthed satellite dataset across 50/54 African countries; trained DINO (ViT backbone) model to 97% accuracy (<i>supervised by Dr. Ka Leung Lam</i>). <i>SpeedTransformer</i>: SOTA 96% Top-1 on Geolife transport-mode prediction (+3%); launched <i>CarbonClever</i>, a social app turning mobility insights into personal CO₂ reductions (<i>supervised by Dr. Charles Chang</i>).	Jun. 2024 – Aug 2025. <i>Suzhou, China</i>
Undergraduate Research Assistant <i>University of Missouri</i> Built a Perceiver IO-based foundation model and a VAE simulator for wildfire spread using satellite + climate data, capturing long-range spatiotemporal dynamics and enabling scenario analysis (<i>supervised by Dr. Jordan Malof</i>).	Jan. 2025 – Jun. 2025 <i>Columbia, MO</i>
Undergraduate Research Fellow <i>Duke University</i> - Mapped invasive salt-marsh species from Sentinel-2/Landsat using ViTs; delivered landscape-scale maps presented at Duke Climate+ 2024 to inform coastal conservation policy (<i>Supervised by Dr. Wenhong Li & Dr. Ding Ma</i>). - Built a geospatial database on saltwater intrusion sea-level rise (North American Coastal Plain); mined 1,000+ papers with BERT; shipped an interactive ArcGIS dashboard (<i>Supervised by Dr. Emily Bernhardt</i>).	Jun. 2023 – Aug. 2024 <i>Durham, NC</i>

PROFESSIONAL EXPERIENCE

Data Analyst Intern <i>Atos Morocco</i> Built real-time HR dashboards used by leadership; scaled deployment across all African branches to standardize decision-making.	Oct. 2022 – Nov. 2022 <i>Rabat, Morocco</i>
Data Analyst Intern <i>XPerlean</i> Applied Faster R-CNN and YOLO for ceramic defect detection, raising accuracy 70%→85% and boosting production speed 10% while reducing quality-control costs.	Jul. 2022 – Aug. 2022 <i>Saint-Quentin, France</i>
Data Analyst Intern <i>Al Jazeera Media Institute</i> Scraped and processed 200k+ social media records via APIs to analyze user behavior and engagement for reporting.	Oct. 2021 – Dec. 2021 <i>Doha, Qatar</i>

PUBLICATIONS & PREPRINTS

- O. Echchabi, A. Lahlou, N. Talty, J. Manto, K. L. Lam. *Tracking Progress Towards Sustainable Development Goal 6 Using Satellite Imagery*. arXiv:2411.19093
- Y. Zhang*, O. Echchabi*, T. Feng, W. Zhang, H.-K. Liao, Z. Lu, C. Chang. *SpeedTransformer: Predicting Human Mobility Patterns Using Dense Smartphone GPS Trajectories and Transformer Models*. International Journal of Geographical Information Science (IJGIS), under revision, 2025.

SELECTED PRESENTATIONS

Monitoring *Spartina alterniflora* Using Self-Supervised Learning. **Duke Climate+ Symposium, 2024.**
Assessing Climate Change Risk of Rural Coastal Plains. **Duke Climate+ Symposium, 2023.**

ACTIVITIES & VOLUNTEER WORK

Duke Kunshan University: Resident Assistant; Student Athlete & Soccer Varsity Co-Captain; CS Club Software Developer [website]; Math & CS Teaching Assistant
FIFA World Cup 2022: Team Leader, Spectator Services Volunteer **FADI Academy:** Partner & Math Tutor

TECHNICAL SKILLS

Languages: Java, Python, C/C++, SQL (Postgres), JavaScript, HTML/CSS, R
Frameworks: React, Node.js, Flask, JUnit, WordPress, Material-UI, FastAPI
Developer Tools: Git, Docker, Google Cloud Platform, VS Code, Visual Studio, PyCharm
Libraries: pandas, NumPy, Matplotlib, PyTorch, TensorFlow, Scikit-Learn

LANGUAGES

Arabic (native); French (fluent); English (fluent); Chinese (intermediate); Spanish (conversational)